

TrES-3b

R-2760

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_110704 · created 2026-09-03T00:20:16+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2455746.8051 +/- 0.00096 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.195 +/- 0.056
Transit depth (Rp/Rs)^2	3.81 +/- 2.19 [%]
Orbital Inclination (inc)	86.35 +/- 2.86
Airmass coefficient 1 (a1)	2.11 +/- 1.18
Airmass coefficient 2 (a2)	-0.67 +/- 1.25
Scatter in the residuals of the lightcurve fit is	57.13 %
Best Comparison Star	#3 - [484, 99]
Optimal Aperture	1.7
Optimal Annulus	4.0
Transit Duration (day)	0.078 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.195 ± 0.056 vs 0.16309 (deviation 0.57σ)
Duration fit vs published	0.078 d vs 0.0567 d (deviation 1.64σ)
Mid-time O–C	1.3 min
Depth SNR	3.5
Residual scatter	57.13 %

Light curve

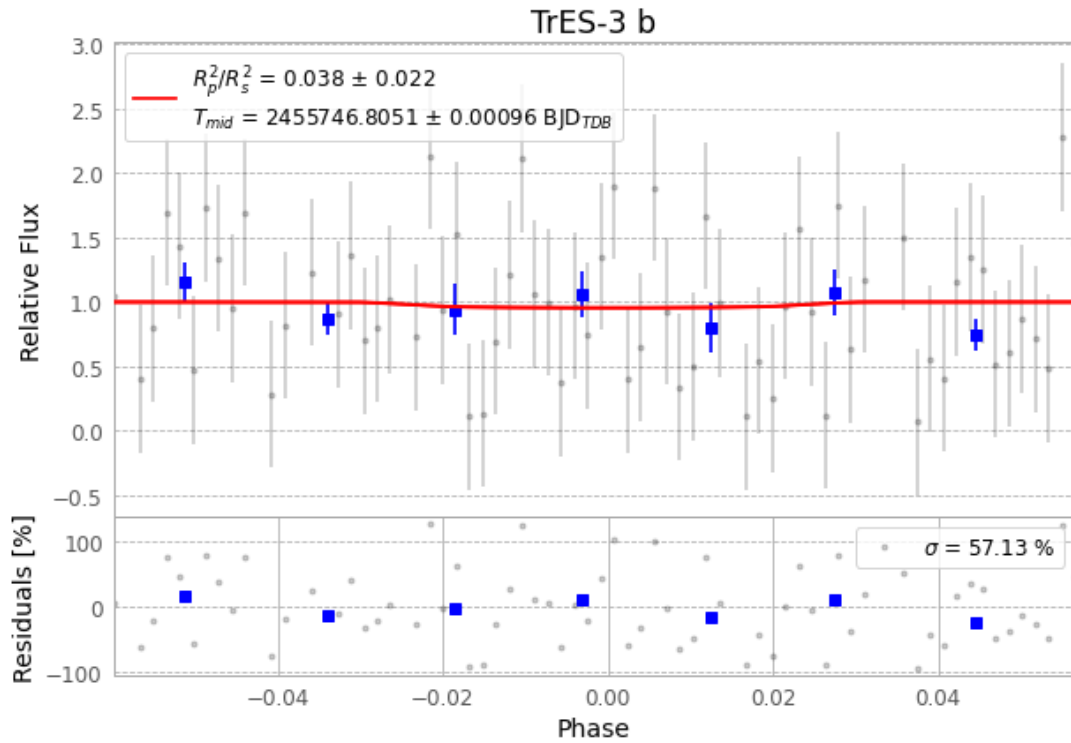


Figure 1 — FinalLightCurve_TrES-3 b_2011-07-03.png (SHA-256 3240890bd3b2a72b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [295, 200], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b42bc50690bc15b1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

74 FITS frames (49 MB), TRES-3110704052114.FITS → TRES-3110704090013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-110704.log (94eeda5f4fcb...), FinalLightCurve_TrES-3 b_2011-07-03.png (3240890bd3b2...), FinalLightCurve_TrES-3 b_2011-07-03.csv (965cf497c002...)

Compute resources

Wall time 211.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-03 00:20Z.